

Giung Lee

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Education

Rice University, George R. Brown School of Engineering

- Bachelor of Science in Computer Science, Minor in Statistics
- Anticipated Graduation Date: May 2027
- Cumulative GPA: 3.77/4.0

Research Interests

Robot safety, safe learning-based systems, visual navigation, and risk-aware planning. I am interested in developing robots that can navigate and manipulate safely in real-world environments by leveraging learned perception, semantic understanding, and environment structure. My research focuses on identifying and reducing safety blind spots that lead to failures in decision-making and execution, with a long-term interest in multi-robot systems where heterogeneous robots must safely coordinate, interact, and share space.

Publications & Manuscripts

Under Review

- Geonyeong Ko^{*}, **Giung Lee^{*}**, and Changjoo Nam. *SAFER-Nav: Enhancing Safety for Visual Robot Navigation via Segmentation-Aware Fine-Tuning*. Under review, Conference on Robot Learning (CoRL), 2026.

In Preparation

- **Giung Lee^{*}**, Weihang Guo^{*}, and Lydia E. Kavraki. *Foundation-Model-Guided Topology-Aware Semantic Risk Fields for Safe Robot Manipulation*. Manuscript in preparation for ICRA 2027.

^{*}Equal contribution.

Relevant Coursework

- **Computer Science:** Algorithmic Robotics, Introduction to Modern Robotics, Reasoning About Algorithms, Practical Machine Learning for Real-World Applications, Computer Architecture, Programming Languages and Compilers, Introduction to Program Design, Data Structures and Algorithms
- **Math:** Multivariable Calculus, Linear Algebra, Applied Multivariate Statistical Method, Probability and Statistics

Experience

Undergraduate Researcher | Kavraki Lab, Rice Univ. – Houston, TX

Jan 2026 – Present

- Developed a **topology-aware semantic risk field** framework for safe robot manipulation, constructing dense **3D risk voxel maps** from RGB-D observations, open-vocabulary perception, LLM-derived object-interaction priors, and voxelized occupancy.
- Designed a reusable JSON database of **directional semantic priors** for manipulated-object / scene-object pairs, using six axis-aligned consequence weights and a spatial spread parameter σ to represent asymmetric hazards beyond scalar pairwise risk scores.
- Implemented **topology-aware risk propagation** by comparing geodesic and Euclidean distance fields, attenuating semantic risk behind shelves, walls, and other occluding geometry that block direct access to scene objects.
- Integrated MobileSAM and SigLIP2-based **open-vocabulary segmentation and 3D grounding** to identify task-relevant scene objects and lift 2D masks into voxelized workspace geometry.
- Demonstrated **end-to-end UR5 manipulation in MuJoCo**, using the semantic risk field for **risk-weighted 3D A* Cartesian planning**, trajectory processing, and Jacobian-based inverse kinematics.
- Showed **manipulated-object-dependent planning behavior**, where changing only the object held by the robot reshapes the risk field and reroutes the planned trajectory under the same scene geometry.

Undergraduate Researcher | AI Robotics Lab, Sogang Univ. – Seoul, South Korea / Remote June 2025 – May 2026

- Co-authored *SAFER-Nav: Enhancing Safety for Visual Robot Navigation via Segmentation-Aware Fine-Tuning* (**under review**, CoRL 2026), proposing a segmentation-aware policy adaptation framework that augments a frozen ViNT backbone with a trainable segmentation encoder, residual attention bridge, and dynamic action blending.
- Extended ViNT into a **37.2M-parameter multimodal navigation model** that fuses 6-frame RGB observations, binary traversability segmentation, and goal images through transformer-based latent refinement and an auxiliary safety-oriented waypoint branch.
- Built a binary traversability pipeline using OneFormer-generated masks and trained new modules with AdamW on **~145k samples from 576 indoor HuRoN trajectories**, while keeping the RGB observation encoder, goal-fusion encoder, and RGB transformer frozen.
- Deployed and evaluated SAFER-Nav in a real-time ROS 2 navigation stack with online segmentation and closed-loop goal-conditioned control on **3 real robot platforms** (RoboMaster S1, TurtleBot4, LoCoBot) across **2 unseen indoor environments** under both static and dynamic obstacle scenarios, benchmarking against ViNT, NoMaD, ViNT+CARE, and NoMaD+CARE.
- Demonstrated consistent safety-performance gains in real-world trials, achieving up to 100% goal arrival with **0.0–0.1 collisions per run** in static obstacle settings and **0/10 collisions** in dynamic-obstacle scenarios.
- Performed mechanism-level ablations showing that reliable obstacle avoidance emerges only when both representation-level direct fusion and the segmentation-only auxiliary action branch are used together.

Teaching Assistant, COMP 215: Intro to Program Design | Rice Computer Science Department – Houston, TX Aug 2025 – May 2026

- Mentored 120+ students in Java, object-oriented programming, data structures, and software design principles.
- Held weekly office hours and review sessions; graded programming assignments using standardized rubrics and delivered constructive feedback.

Sergeant, UAS Mission Commander | U.S. Army KATUSA, Engineering Battalion – Camp Hovey, South Korea Nov 2022 – May 2024

- Served as a squad leader in the UAS platoon, leading soldiers during daily duty operations, morning physical training, and platoon-level coordination.
- Supervised 150+ UAV flight operations as UAS Mission Commander, coordinating pre-flight safety checks, mission execution, and operational readiness.
- Acted as a military liaison between the U.S. Army and the ROK Army, supporting safe UAV flight training in Korean airspace and compliance with both ROK and U.S. military aviation standards.

Projects

SPACE-K | UAV Business Competition Project Mar 2023 – Apr 2024

- Led a student team as CEO in developing a business concept for a low-altitude UAV tracking and communication system using LTE and SATCOM technologies.
- Produced a 10+ page business plan and delivered investor-style presentations across three competition stages: market analysis, prototyping, and pitching.
- Advanced the project through multiple competition stages by refining the technical concept, business model, and deployment scenario for UAV operations.

Service

Korea Connect | Rice Alzheimer's Buddies – Houston, TX Nov 2024 – Present

- Lead small-group sessions connecting Rice students with Korean seniors at risk for social isolation and dementia through bilingual companionship and activity-based engagement.
- Facilitate Korean–English communication between students, seniors, and community partners, supporting translation during group activities and event coordination.
- Help organize semesterly cultural engagement events, including creative expression, physical activities, guest lectures, games, refreshments, and campus tours.

Honors & Awards

- **Korea Presidential Science Scholarship** | 1 of 20 students selected nationwide
- **Rice University President's Honor Roll** | Fall 2025
- **1st Place** | 2024 KT&G Alliance Students' Venture Forum
- **Creativity Award** | Army Start-up Business Competition
- **Certificate of Appreciation** | 14th Brigade Engineer Battalion
- **Judges' Award** | Maryland Tech Invitational, Team Shatterdome, 2019

Additional Skills

- **Programming:** Python, C/C++, Java, JavaScript
- **Robotics & Machine Learning:** PyTorch, ROS 2, MuJoCo, OpenCV, scikit-fmm, NumPy, Pandas
- **Tools:** Git/GitHub, Linux, LaTeX
- **Languages:** Korean, English